Automated Classification of Blood Cells from Peripheral Blood Smear Images

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Contents

Abstract

This project presents a complete machine learning pipeline for the automated classification of blood cells into eight distinct categories from peripheral blood smear microscopic images. The pipeline implements a classical image-processing approach: nucleus segmentation using a novel combination of the CMYK and HLS colour spaces, followed by manual extraction of 51 morphological and colour features, rigorous feature validation and reduction to 19 features, and finally a systematic comparison of four classifiers Support Vector Machine (SVM), Random Forest (RF), k-Nearest Neighbours (KNN), and a soft-vote ensemble. The best-performing model, an RBF-kernel SVM with tuned hyperparameters (C = 51,526.95, $\gamma = 0.00101$), achieved a 5-fold cross-validation accuracy of 93.53% ± 0.58% and a held-out test accuracy of 92.26% with a macro F1-score of 89.86% across all eight classes including the clinically challenging immature granulocyte (IG) and monocyte categories.

1 Introduction

The differential counting of blood cells is a routine haematological procedure that provides critical diagnostic information for a wide range of conditions including leukaemia, anaemia, viral infections, and immune disorders. In clinical practice, this analysis is typically performed manually by trained haematologists examining Giemsa-stained blood smears under a microscope, a process that is time-consuming, subject to inter-observer variability, and dependent on the availability of specialist expertise.

Automated classification systems offer the potential to make this process faster, more consistent, and accessible in resource-limited settings. However, the task is non-trivial: different blood cells types can be visually similar, particularly between morphologically related classes such as immature granulocytes and other granulocyte subtypes, and imaging conditions such as staining technique, microscope type, and camera quality introduce additional variability.

This project aims to build a reproducible, interpretable, and lightweight classification pipeline that avoids the computational expense of deep learning while achieving high classification accuracy across eight WBC categories. The approach is grounded in classical machine learning: handcrafted morphological and colour features are extracted from segmented nuclei and cytoplasm regions, validated against biological expectations, and used to train and compare multiple classifiers.

The eight classes targeted in this project are: basophil, eosinophil, erythroblast, immature granulocyte (IG), lymphocyte, monocyte, neutrophil, and platelet.

2 Related Work

The classification of blood cells has been studied extensively. Early works relied on thresholding algorithms such as Otsu's method for nucleus segmentation followed by shape features computed on the resulting binary masks. Rezatofighi and Soltanian-Zadeh (2011) combined

Gram-Schmidt decomposition for nucleus extraction with local binary patterns and grey-level co-occurrence matrices for feature extraction. Hiremath et al. (2010) used global thresholding with geometric nucleus and cytoplasm features.

More recent work has shifted towards deep learning. Jung et al. (2019) proposed W-Net, a CNN architecture specifically designed for WBC classification, while Baydilli and Atila (2020) applied capsule networks to the same problem. Pre-trained CNN models including ResNet, AlexNet, and MobileNet have been fine-tuned for blood cells classification with strong results on individual datasets.

Tavakoli et al. (2021) demonstrated that a carefully engineered classical feature set — combining CMYK/HLS-based nucleus segmentation with handcrafted shape and colour features fed into an SVM — could match or outperform pre-trained CNN models. The segmentation and feature extraction methodology used in this project directly extends that work to an eight-class problem.

3 Dataset

The raw dataset consists of 17,092 original images spanning eight blood cells classes collected from Giemsa-stained peripheral blood smear microscopy images. The class distribution in the original data is naturally imbalanced, with neutrophils being the most abundant class and lymphocytes and basophils the rarest, reflecting the physiological proportions of these cells in normal peripheral blood.

Class	Original Count
neutrophil	3,329
eosinophil	3,117
IG (immature granulocyte)	2,895
platelet	2,348
erythroblast	1,551
monocyte	1,420
basophil	1,218
lymphocyte	1,214
Total	**17,092**

Table 1: Original dataset class distribution.

Each image is a cropped RGB photograph centred on a single white blood cell, stored as a JPEG file. Images vary in resolution and background colour depending on the source microscope and camera used during acquisition.

4 Methodology

The pipeline is implemented across five sequential notebooks, each corresponding to a distinct stage of the analysis. The overall workflow is illustrated below:

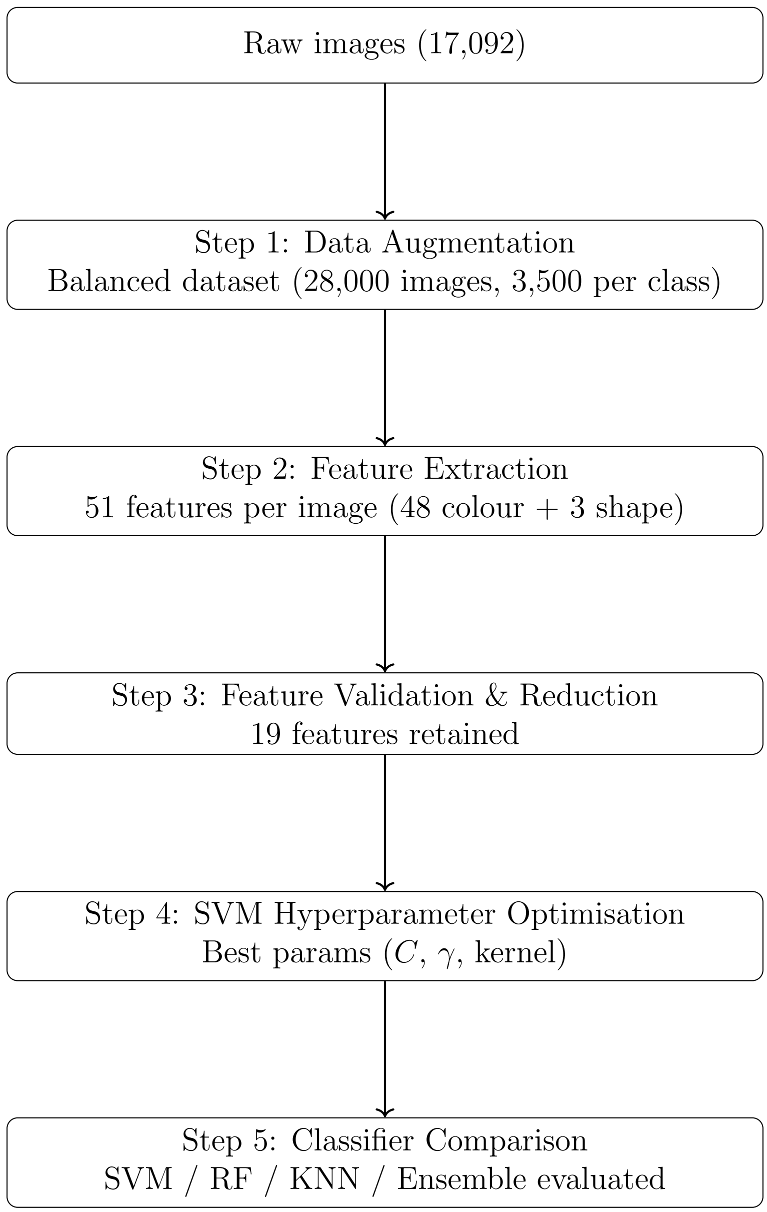


Figure 1: Overview of the blood cells classification pipeline

4.1 Data Augmentation

To address the class imbalance in the original dataset, a rotation-based augmentation strategy was applied using a custom ‘AugmentImageDataSet‘ class. Each class was upsampled to exactly 3,500 images by generating rotated copies — with rotations drawn uniformly from the range 60 to +60 of original images that were selected as augmentation sources.

The augmentation produced 10,908 new images distributed as follows:

Class	**Original**	**Augmented**	**Total**
Basophil	1,218	2,282	3,500
Eosinophil	3,117	383	3,500
Erythroblast	1,551	1,949	3,500
IG	2,895	605	3,500
Lymphocyte	1,214	2,286	3,500
Monocyte	1,420	2,080	3,500
Neutrophil	3,329	171	3,500
Platelet	2,348	1,152	3,500
Total	**17,092**	**10,908**	**28,000**

Table 2: Data Augmentation Summary

The rotation range of 60 was chosen to generate realistic morphological variation while avoiding extreme orientations that would produce implausible imaging artefacts. Because blood cell nuclei are not orientation-dependent structures, rotated copies are biologically equivalent to the originals and constitute valid training examples.

A critical design decision was made to track augmentation provenance for every image using an 'augmented' flag column (0 = untouched original, 1 = original used as rotation source, 2 = generated rotation). This flag was used throughout the pipeline to ensure that no augmented images contaminated the test set, preventing data leakage.

4.2 Image Segmentation and Feature Extraction

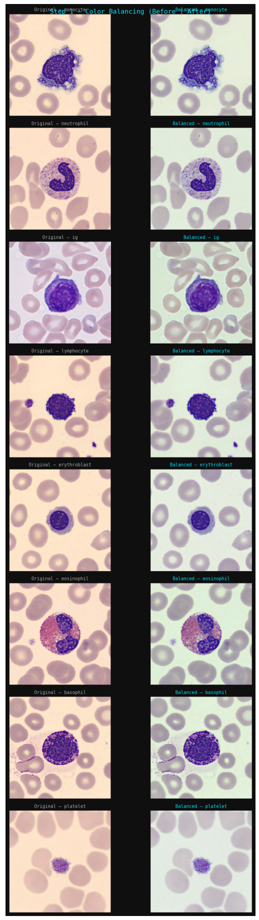
Feature extraction followed the methodology of Tavakoli et al. (2021), extended to eight classes. For each image, the pipeline performed three steps: nucleus segmentation, cytoplasm region detection, and feature computation. The objective was to transform each microscopy image into a 51-dimensional feature vector $f \in \mathbb{R}^{51}$ encoding geometric morphology and chromatic distribution.

4.2.1 Step 1 — Grey-World Colour Normalisation

Before any colour-space computation, illumination bias was removed using grey-world normalisation. The spatial average of reflectance is assumed achromatic, meaning that the average colour of a scene under neutral illumination should be grey. Each channel of the RGB image — where R (raw red light intensity), G (raw green light intensity), and B (raw blue light intensity) — was scaled so that its mean matched the mean of the greyscale representation:

$$I'_c = I_c \cdot \frac{\mu_{\text{gray}}}{\mu_c}, \quad \mu_c = \frac{1}{N} \sum_{x,y} I_c(x, y) \tag{1}$$

where I_c is the input intensity matrix for channel $c \in \{R, G, B\}$, I'_c is the normalised output, μ_c is the per-channel mean, μ_{gray} is the mean intensity of the greyscale image, N is the total number of pixels, and x, y are spatial coordinates. This step reduces staining variability introduced by different laboratories and imaging conditions.



4.2.2 Step 2 — Nucleus-Enhancing Channel Construction

The colour-balanced RGB image was converted simultaneously into two specialist colour spaces to construct a nonlinear fusion channel that emphasises chromatin density.
CMYK conversion describes colour as layers of printer ink: C (cyan), M (magenta), Y (yellow), and K (key/black). Under Giemsa staining, the nucleus absorbs the magenta dye strongly but requires very little black ink, while red blood cells require both M and K in similar proportions. The magenta channel M encodes the pink-purple dye concentration, and the black channel K encodes overall ink density.

HLS conversion describes colour in a perceptually intuitive way: H is the hue (the dominant colour angle on the colour wheel, distinguishing purple from pink), L is the lightness (perceptual brightness), and S is the saturation (colour purity; 0 = grey, 1 = vivid colour). The nucleus is highly saturated under Giemsa staining, while the plain white background has near-zero saturation.

Three intermediate quantities were computed from these two colour spaces:

$$\text{min\_MS} = \min(M, S) \tag{2}$$
$$KM = K - \min(K, M) \tag{3}$$
$$\text{Soft map} = \text{min\_MS} - \min(\text{min\_MS},\ KM) \tag{4}$$

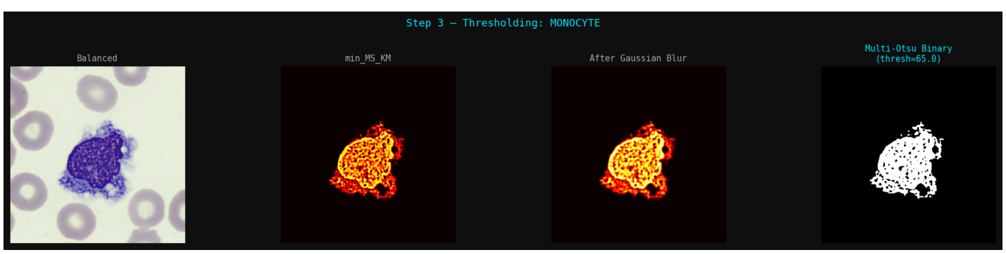
The first quantity, min_MS, takes the minimum of the magenta channel M and the saturation channel S (colour purity from HLS) pixel-by-pixel. This is bright wherever both M and S are high — namely the nucleus — and near-zero where either is low, which suppresses the featureless white background. The second quantity, KM, is high wherever black ink (K) dominates magenta (M) — i.e., the red blood cells — and near-zero at the nucleus where M dominates K. The final soft map subtracts KM from min_MS: red blood cells are bright in both and cancel to zero; background is dark in both and also cancels; but the nucleus is bright in min_MS and dark in KM, so it survives as the dominant bright structure.

4.2.3 Step 3 — Gaussian Smoothing

The soft map was convolved with a Gaussian kernel prior to thresholding to suppress high-frequency pixel noise:

$$I_{\text{blur}}(x, y) = \big(I * G_\sigma\big)(x, y), \quad G_\sigma(x, y) = \frac{1}{2\pi\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{5}$$

where $*$ denotes convolution, G_σ is the Gaussian kernel with standard deviation σ controlling the extent of smoothing, and x, y are spatial coordinates. A kernel size of 5×5 pixels was used.



4.2.4 Step 4 — Multi-Class Otsu Segmentation

A 3-class multi-Otsu threshold was applied to the blurred soft map, partitioning intensity into three classes by minimising the weighted intra-class variance:

$$\sigma_w^2 = \sum_{k=1}^{3} \omega_k \sigma_k^2 \tag{6}$$

where ω_k is the probability weight of class k and σ_k^2 is its variance. The algorithm returns two thresholds t_1 and t_2. Only the highest-intensity class — the nucleus — was retained as a binary mask:

$$N(x,y) = \begin{cases} 1 & I(x,y) \geq t_2 \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

This three-class approach was essential for correctly separating nucleus from cytoplasm: a two-class threshold collapsed cytoplasm and nucleus into a single class for monocyte and IG images.

4.2.5 Step 5 — Largest Connected Component Selection

Among all contours detected in the binary mask, only the single largest was retained:

$$C^* = \arg\max_{C_i} \; \text{Area}(C_i) \tag{8}$$

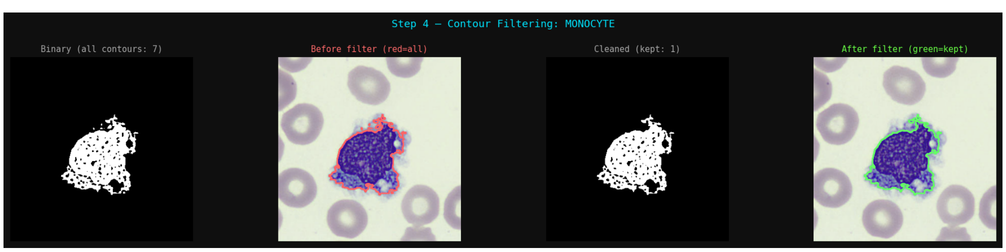
where C_i denotes each individual detected contour and $\text{Area}(C_i)$ is its enclosed area. This eliminates small debris from neighbouring cells, staining artefacts, and fragmented partial segmentations.

4.2.6 Step 6 — Morphological Closing and Hole Filling

The selected contour mask was regularised topologically using morphological closing:

$$A \bullet B = (A \oplus B) \ominus B$$

where A is the binary nucleus mask, B is an elliptical structuring element, $\oplus$ is dilation, $\ominus$ is erosion, and $\bullet$ is the closing operator. This sealed hairline cracks caused by chromatin texture. A subsequent flood-fill from the image border identified the true external background, and inverting the result filled all remaining interior voids, producing a topologically solid nucleus mask.



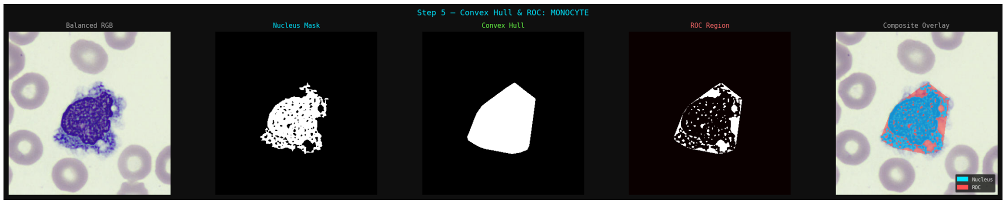
4.2.7 Step 7 — Convex Hull and Region Definitions

Three regions were defined from the final binary nucleus mask N, following the notation of the feature annotation guide:

$$S = \{(x,y) : N(x,y) = 1\}, \quad H = \text{Conv}(S)$$

$$\text{NCL (Nucleus)} = S, \quad \text{CVX (Convex Hull)} = H, \quad \text{ROC (Ring of Cytoplasm)} = H \setminus S \quad (9)$$

where S is the set of nucleus pixels, H is the smallest convex polygon enclosing S (the convex hull), and $\setminus$ is the set difference operator. The NCL region contains the segmented nuclear pixels. The CVX region is the convex hull and serves as a normalisation denominator to correct for illumination variation across images. The ROC region — the hull minus the nucleus — captures the cytoplasm ring immediately surrounding the nucleus, providing a cytoplasm-representative region without requiring full cytoplasm segmentation.

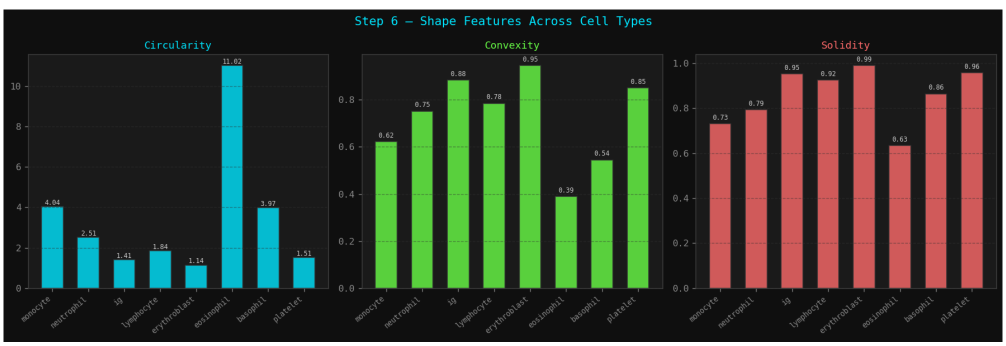


4.2.8 Step 8 — Shape Descriptors

Three shape features were computed from the nucleus contour geometry:

$$\text{Circularity} = \frac{P_{smooth}^2}{4\pi A}, \quad \text{Convexity} = \frac{P_{hull}}{P_{nucleus}}, \quad \text{Solidity} = \frac{A_{nucleus}}{A_{hull}} \quad (10)$$

where P_{smooth} is the Gaussian-smoothed nucleus perimeter (smoothed using a 1D Gaussian filter with $\sigma = 1.5$, applied with wrap-around boundary conditions to the contour coordinates to suppress rasterisation noise), A is the nucleus area, P_{hull} is the perimeter of the convex hull polygon, $P_{nucleus}$ is the raw nucleus perimeter, $A_{nucleus}$ is the nucleus pixel count, and A_{hull} is the convex hull area. Circularity equals 1.0 for a perfect circle and increases without bound as the shape becomes more irregular. Convexity and Solidity both lie in (0, 1], with 1.0 indicating a perfectly convex, gap-free nucleus.



4.2.9 Step 9 — Colour Channels and Statistical Moments

Colour information was extracted from four colour spaces, providing 12 channels in total:

RGB: R (raw red light intensity), G (raw green light intensity), B (raw blue light intensity — the dominant staining channel for nuclei under Giemsa)

HSV: H (hue — dominant colour angle on the colour wheel, distinguishing purple nucleus from pink cytoplasm), S (saturation — colour purity; 0 = grey, 1 = vivid colour), V

(value/brightness — maximum of R, G, B).
CIE Lab: L (perceptual lightness, fully decoupled from colour), A (the a green-to-red opponent axis; negative = green, positive = red), Bstar (the b blue-to-yellow opponent axis; negative = blue, positive = yellow — nuclei staining blue-purple are expected to yield negative values here).
YCrCb: Y (luma — brightness approximation, near-identical to L), Cr (red-difference chroma — how much redder than neutral grey), Cb (blue-difference chroma — how much bluer than neutral grey).

For each of the three regions (NCL, ROC, CVX) and each of the 12 channels, two statistical moments were computed:

$$\mu_{R,c} = \frac{1}{|R|} \sum_{(x,y)\in R} I_c(x,y), \quad \sigma_{R,c} = \sqrt{\frac{1}{|R|} \sum_{(x,y)\in R} (I_c(x,y) - \mu_{R,c})^2} \tag{11}$$

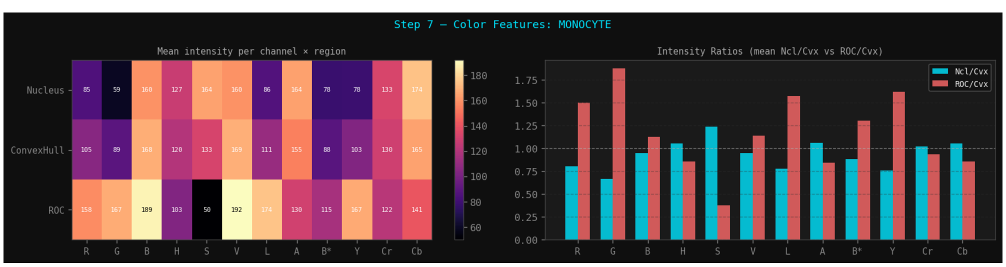
where $R \in \{\mathrm{NCL}, \mathrm{ROC}, \mathrm{CVX}\}$ is the region of interest, c is the colour channel index, $|R|$ is the pixel count of the region, $I_c(x,y)$ is the intensity at coordinate (x,y) for channel c, $\mu_{R,c}$ is the region mean, and $\sigma_{R,c}$ is the region standard deviation.

4.2.10 Step 10 — Ratio Normalisation

Rather than using absolute intensity statistics, each region's mean and standard deviation were expressed as ratios relative to the corresponding CVX (convex hull) statistic. The convex hull region serves as a local illumination reference:

$$R^{mean}_{NCL,c} = \frac{\mu_{NCL,c}}{\mu_{CVX,c}}, \quad R^{std}_{ROC,c} = \frac{\sigma_{ROC,c}}{\sigma_{CVX,c}} \tag{12}$$

A small constant ϵ was added to all denominators to prevent division by zero. This ratio normalisation enforces illumination invariance: two images of the same cell type taken under different illumination conditions will produce similar ratio values even if their absolute intensities differ substantially.



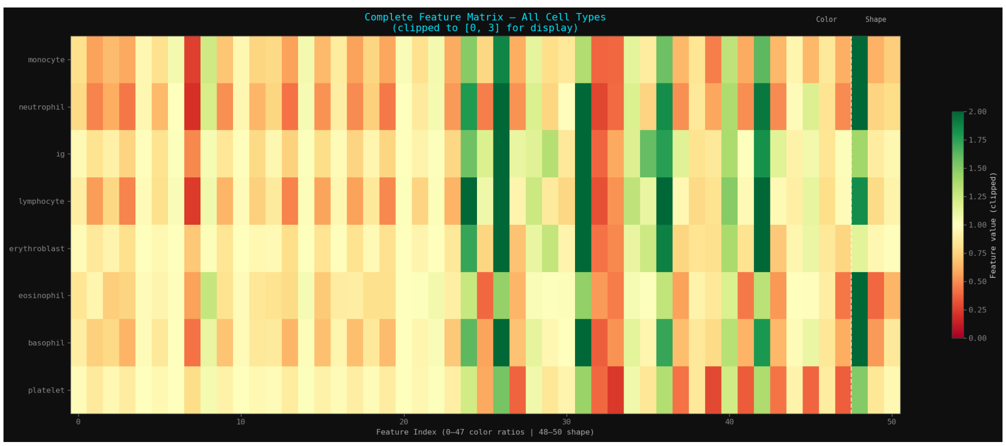
4.2.11 Final Feature Vector

Concatenating all descriptors produces the 51-dimensional feature vector $f \in \mathbb{R}^{51}$:

- 12 NCL/CVX mean ratios — average colour of nucleus relative to whole-cell average, across all 12 channels.

- 12 NCL/CVX std ratios — internal colour texture of nucleus relative to whole-cell spread.
- 12 ROC/CVX mean ratios — average colour of the cytoplasm ring relative to whole-cell average.
- 12 ROC/CVX std ratios — cytoplasm ring colour texture relative to whole-cell spread.
- 3 shape descriptors — Circularity, Convexity, Solidity.

Extraction was parallelised across 16 CPU workers using Python's multiprocessing module, with checkpoint-resume capability to survive interruptions. The full dataset of 28,000 images was processed with a checkpoint written every 500 rows.



4.3 Feature Validation and Reduction

Before training any classifier, the 51-feature set was subjected to a four-stage validation and reduction pipeline.

4.3.1 Check 1 — Mathematical Validity

Each feature was verified against its geometric definition: Circularity ≥ 1.0, Convexity $\in (0, 1]$, Solidity $\in (0, 1]$, and all colour ratios non-negative and finite.

Initial extraction produced 65 Circularity violations and 10 NaN rows, both confined to the platelet class. Platelets have no nucleus; the segmentation pipeline processes the platelet cell body instead. Because the platelet body is nearly circular, Gaussian smoothing of the contour perimeter and flood-fill hole-filling reduce the $\frac{P^2}{4\pi A}$ ratio fractionally below 1.0. The 65 Circularity values were clipped to 1.0.

The 10 NaN rows arose because the 3-class multi-Otsu thresholding assigned the platelet cell body to the mid-intensity class rather than the top class, producing an empty binary mask and no extractable features. These 10 rows (0.036% of the dataset) were dropped.

After corrections, all 27,990 remaining rows passed all mathematical checks.

4.3.2 Check 2 — Biological Expectations

Per-class mean values for the three shape features were compared against expected ranges from haematology literature. The key biological ordering — lymphocytes should have the lowest circularity while neutrophils should have the highest — was confirmed using Mann-Whitney U tests. All three shape features yielded p-values indistinguishable from zero between lymphocytes and neutrophils.

Class	Circularity	Convexity	Solidity
erythroblast	1.1467	0.9411	0.9899
lymphocyte	1.1921	0.9312	0.9816
ig	1.5037	0.8809	0.9283
platelet	1.5870	0.8848	0.8832
eosinophil	1.7541	0.8687	0.8808
basophil	1.8194	0.8184	0.8785
neutrophil	2.0271	0.8251	0.8442
monocyte	2.1955	0.8287	0.8514

Table 3: Per-class mean shape features

Per-class mean shape features The monocyte Circularity mean (2.20) slightly exceeded its expected range (1.10–1.60), consistent with the known variability of monocyte nuclear morphology. This was logged as a warning but did not disqualify the feature.

4.3.3 Check 3 — Inter-class Separability (ANOVA F-test)

A one-way ANOVA F-test was applied to each feature to quantify its discriminative power across the eight classes. All 51 features achieved an F-score above 500, with p-values effectively zero.

The highest-scoring features were dominated by colour ratio features from the ROC region. The top features included:

- RocCvx_Cb_mean (Cb: blue-difference chroma, YCrCb) — F = 5615
- RocCvx_Bstar_mean (b: blue-yellow axis, CIE L*a*b*) — F = 4578
- RocCvx_Cr_mean (Cr: red-difference chroma, YCrCb) — F = 4429
- NclCvx_R_mean (R: red channel, RGB) — F = 4088

This confirms that cytoplasm colour contrast — particularly blue and red chroma — is the dominant discriminative signal.

4.3.4 Correlation-Based Redundancy Removal

Pairwise Pearson correlations were computed across all 51 features. Within each pair exceeding the $|r| \geq 0.90$ threshold, the feature with the lower F-score was removed. The three shape features were protected from removal.

This eliminated 32 redundant features, leaving 19 features. No additional features were removed by the F-score threshold.

Feature	Type	Colour Meaning	F-score
RocCvx_Cb_mean	Colour (ROC mean)	Cb (YCrCb)	5615.5
RocCvx_Cr_mean	Colour (ROC mean)	Cr (YCrCb)	4429.0
NclCvx_R_mean	Colour (NCL mean)	R (RGB)	4088.5
RocCvx_B_mean	Colour (ROC mean)	B (RGB)	3753.7
RocCvx_A_std	Colour (ROC std)	a* (Lab)	3473.0
NclCvx_B_std	Colour (NCL std)	B (RGB)	3391.4
RocCvx_R_mean	Colour (ROC mean)	R (RGB)	3583.0
RocCvx_G_std	Colour (ROC std)	G (RGB)	2731.6
RocCvx_Cr_std	Colour (ROC std)	Cr (YCrCb)	2490.4
RocCvx_B_std	Colour (ROC std)	B (RGB)	2224.0
RocCvx_H_mean	Colour (ROC mean)	H (HSV)	2015.7
NclCvx_H_std	Colour (NCL std)	H (HSV)	1958.1
Convexity	Shape	–	1684.5
Solidity	Shape	–	1311.5
NclCvx_H_mean	Colour (NCL mean)	H (HSV)	1231.2
NclCvx_Cr_mean	Colour (NCL mean)	Cr (YCrCb)	2629.4
NclCvx_Cr_std	Colour (NCL std)	Cr (YCrCb)	1026.2
RocCvx_H_std	Colour (ROC std)	H (HSV)	788.0
Circularity	Shape	–	738.5

Table 4: Final selected features after redundancy removal

Final retained features A benchmark SVM (C=10, γ = scale, RBF kernel) achieved 93.15% cross-validated accuracy using the 19-feature set, compared to 94.35% using all 51 features, a reduction of only 1.2%. This confirms that the reduced feature set retains the majority of the discriminative signal while significantly reducing dimensionality.

4.4 Hyperparameter Optimisation

Hyperparameter search was performed exclusively on the SVM with an RBF kernel, using original-only images for both training and evaluation to prevent augmented data from inflating estimates. A leakage-safe data split was used throughout: the 20% test set was drawn entirely from untouched original images (augmented flag = 0), and all augmented rows were added to the training pool only.

4.4.1 Stage 1 — Coarse RandomizedSearchCV

A random search over 60 candidate configurations was performed using 5-fold stratified cross-validation. The search space was defined using log-uniform distributions: $C \in [0.1, 10, 000]$ and $\gamma \in [0.00001, 10]$, with kernel $\in \{rbf, poly\}$. The best configuration found was C = 8,244.31, $\gamma = 0.00632$, kernel = rbf, yielding a CV accuracy of 91.93%.

4.4.2 Stage 2 — Fine GridSearchCV

A dense grid centred on the Stage 1 optimum was constructed using 9 logarithmically spaced values around the best C and γ. The resulting $9 \times 9 = 81$ configurations were exhaustively evaluated via 5-fold CV. The fine grid search identified the optimal configuration as $C = 51, 526.95$, $\gamma = 0.00101$, kernel = rbf, achieving a CV accuracy of 92.11% on original images alone. This configuration was retrained on the full augmented training pool (originals + augmented copies) for final evaluation.

4.5 Classifier Comparison

Four classifiers were trained and compared on the same leakage-safe train/test split, each wrapped in a scikit-learn Pipeline that applied StandardScaler normalisation before the classifier:

- **SVM:** Best tuned parameters from Step 4 (C = 51,526.95, $\gamma = 0.00101$, RBF kernel, balanced class weights, one-vs-rest strategy)
- **Random Forest:** 500 trees, balanced class weights
- **KNN:** 7 neighbours, distance-weighted voting
- **Soft-Vote Ensemble:** Combined SVM + RF + KNN using predicted class probabilities (soft voting)

All models were evaluated using 5-fold stratified cross-validation on the training pool and a final held-out test set of 2,337 original images with a minimum of 100 images per class guaranteed.

5 Experimental Setup

- **Hardware:** MacBook with 16 logical CPU cores. Feature extraction was parallelised across all 16 cores. SVM training was performed on CPU only (no GPU required).
- **Software:** Python 3, scikit-learn, OpenCV (cv2), NumPy, scikit-image (multi-Otsu thresholding), pandas, matplotlib, seaborn.
- **Train/Test split strategy:** The split was performed on original images only (augmented flag = 0) first, guaranteeing that the test set consisted exclusively of untouched originals. All augmented images were then added to the training pool. This two-step

Split	Size	Composition
Training pool	25,653	8,476 originals + 17,177 augmented
Test set	2,337	Originals only (min 100 per class)

Table 5: Train/Test split composition

process ensures zero leakage: no augmented derivative of a test image could appear in training.

- **Evaluation metrics:** 5-fold stratified cross-validated accuracy (mean $\pm$ std), held-out test accuracy, macro-averaged F1-score (unweighted average across all eight classes), and per-class precision, recall, and F1-score.

6 Results

6.1 Feature Extraction Results

Feature extraction was completed for 27,990 of 28,000 images (99.96%). The 10 failed extractions were all from the platelet class and were caused by the absence of a true nucleus, leading to an empty segmentation mask. These rows were dropped prior to training. Following Circularity clipping and NaN removal, all 27,990 rows passed the full validation suite.

6.2 Feature Validation Results

- **Mathematical validity:** All 51 features passed all boundary checks after patching. Circularity values ranged from 1.000 to 18.139 (mean 1.653). Convexity ranged from 0.453 to 1.000 (mean 0.872). Solidity ranged from 0.181 to 1.000 (mean 0.905). No colour ratios were negative, infinite, or NaN.
- **Biological ordering:** The three Mann-Whitney U tests comparing lymphocytes (compact, round nucleus) against neutrophils (multi-lobed nucleus) all produced p-values indistinguishable from zero, confirming that the shape features carry genuine discriminative biological signal:
  - Lymphocyte Circularity (1.197) $<$ Neutrophil Circularity (2.027) — $p \approx 0$
  - Lymphocyte Convexity (0.929) $>$ Neutrophil Convexity (0.825) — $p \approx 0$
  - Lymphocyte Solidity (0.980) $>$ Neutrophil Solidity (0.844) — $p \approx 0$
- **Inter-class separability:** All 51 features had ANOVA F-scores ≥ 500 and p-values effectively zero, confirming that every feature carries significant class-discriminative signal.

6.3 Feature Reduction Results

Stage	Features Remaining
Original feature set	51
After correlation-based removal ($\|r\| \geq 0.90$)	19
After F-score filter ($F \geq 500$)	19 (no additional removal)
Final reduced set	**19**

Table 6: Feature reduction stages

The reduced 19-feature set achieved a cross-validated SVM accuracy of **93.15%** versus **94.35%** for the full 51-feature set — a loss of only 1.2 percentage points while reducing the number of features by 63%.

6.4 Hyperparameter Tuning Results

The fine-grid hyperparameter heatmap (Figure 7) shows the 5-fold cross-validated accuracy across the 9×9 $C \times \gamma$ search space. The optimal region was centred around high C values ($C \approx$ 50,000–130,000) and moderate-to-low γ values ($\gamma \approx 0.001$), consistent with the need for a high-margin, moderately smooth decision boundary. The best single configuration achieved 92.11% cross-validated accuracy on original images alone.

When retrained on the full augmented training pool (25,653 samples), the same hyperparameters achieved **5-fold cross-validated accuracy of** 93.53% $\pm$ 0.58%, with per-fold scores of 92.6%, 93.4%, 93.9%, 94.3%, and 93.4%.

6.5 Model Comparison Results

Model	CV Accuracy (%)	CV Std (%)	Test Accuracy (%)	Macro F1 (%)	Train Time
SVM	**93.53**	**0.58**	**92.26**	**89.86**	2.2 min
Soft-Vote Ensemble	93.21	0.28	91.14	88.19	2.4 min
Random Forest	91.10	0.36	89.13	85.68	0.1 min
KNN	89.30	0.06	88.19	83.77	< 1 s

Table 7: Model performance comparison

The SVM outperformed all other models across every evaluation metric. The soft-vote ensemble ranked second, demonstrating that aggregating class probability distributions from SVM, Random Forest, and KNN yielded a modest improvement over Random Forest and KNN individually, but did not surpass the tuned SVM.

The Random Forest provided the best trade-off between performance and training time. KNN was the fastest to train but produced the lowest accuracy, likely due to the curse of dimensionality in the 19-dimensional feature space.

The 5-fold cross-validation boxplot (Figure 3) further confirms that the SVM achieved the highest median accuracy ($\sim$ 93.5%) with the widest interquartile range, indicating greater sensitivity to fold composition. The ensemble model exhibited the tightest distribution, suggesting more stable predictions across folds. KNN showed the lowest variance but consistently achieved the lowest accuracy.

6.6 Per-Class Performance

The per-class F1-score heatmap (Figure 2) reveals consistent patterns across all four models:

- **Platelet:** Achieved the highest F1-score across all models (SVM: 97.4%), reflecting the morphologically distinct and consistent appearance of platelet cell bodies under staining.
- **Eosinophil:** Highly discriminable (SVM: 96.3%), consistent with its characteristic bilobed nucleus and distinctively granulated cytoplasm, which produce a strong signal in the Cr (red-difference chroma, YCrCb — how much redder than neutral grey) and Bstar (b^* blue-to-yellow opponent axis, CIE L*a*b*) channels.
- **Monocyte:** The hardest class for all models (SVM: 77.3%), reflecting the high morphological variability of monocyte nuclei and their visual overlap with other mononuclear classes in the H (hue — dominant colour angle on the colour wheel) and S (saturation — colour purity) channels.
- **Immature granulocyte (IG):** The second most challenging class (SVM: 87.2%), expected given that IGs represent a spectrum of maturation stages with overlapping morphologies bridging band forms and mature granulocytes.

The confusion matrices for the SVM (Figure 5) confirm these patterns. The largest off-diagonal mass involves IGs being misclassified as basophils (14 cases) and neutrophils (12 cases), which is biologically reasonable — band forms and metamyelocytes overlap morphologically with mature granulocytes. The monocyte row shows the most scattered misclassifications, spread across IG (6 cases) and platelet (3 cases).

Class	Precision	Recall	F1-Score	Support
basophil	0.829	0.920	0.872	100
eosinophil	0.972	0.955	0.963	551
erythroblast	0.843	0.910	0.875	100
ig	0.896	0.849	0.872	469
lymphocyte	0.928	0.900	0.914	100
monocyte	0.677	0.900	0.773	100
neutrophil	0.958	0.935	0.946	631
platelet	0.972	0.976	0.974	286
macro avg	**0.884**	**0.918**	**0.899**	**2,337**
accuracy			**0.923**	2,337

Table 8: SVM classification report on the held-out test set

7 Discussion

The SVM's superiority over the ensemble, Random Forest, and KNN can be attributed to two main factors. First, the feature space is relatively low-dimensional (19 features) and the classes are well-separated by colour ratio features — a regime where SVMs with non-linear kernels excel. Second, the hyperparameter search specifically optimised the SVM's C and γ parameters for this task, while the Random Forest and KNN were used with reasonable but untuned defaults.

The high importance of cytoplasm-region colour features — particularly ratios involving **Cb** (blue-difference chroma from YCrCb — how much bluer than neutral grey) and **Cr** (red-difference chroma — how much redder than neutral grey) in the **ROC** (Ring of Cytoplasm — the cytoplasm ring pixels minus the nucleus) region — reflects a key biological reality: the granule content and cytoplasm staining colour are among the most diagnostically distinctive features that haematologists use to identify WBC types visually. Eosinophil granules stain strongly with eosin (bright orange-red, yielding high Cr values in the ROC), while basophil granules stain intensely with the basic dye (deep blue-purple, yielding low Bstar — b^* blue-to-yellow opponent axis — and low Cb values in the ROC). These colour signatures are well captured by the ratio normalisation approach.

The relatively low performance on the monocyte class (macro F1 of 77.3% for SVM) highlights a fundamental challenge: monocyte nuclear morphology is highly variable, and their large, pale, sometimes vacuolated cytoplasm can resemble other large mononuclear cells. The **H** (hue — dominant colour angle on the colour wheel) channel in both the NCL (nucleus) and ROC (cytoplasm ring) regions provides some discriminative signal, but it is insufficient to reliably separate monocytes from other irregular-nucleus classes.

The feature reduction from 51 to 19 features, with a loss of only 1.2 percentage points, validates the substantial redundancy in the original feature set. The 32 dropped features were all highly correlated ($|r| \geq 0.90$) with a higher-scoring surviving feature, typically because multiple colour spaces encode similar information. For example, **Y** (luma from YCrCb — near-identical to L perceptual lightness from CIE L*a*b*) was dropped due to $r = 0.999$ with L, and **G** (raw green light intensity from RGB) was dropped due to $r = 0.996$ with L, since green light dominates the luminance perception model.

8 Limitations and Future Work

- **Monocyte classification** remains the primary performance bottleneck. Future work could investigate class-specific feature engineering — for example, second-order texture features from the cytoplasm using the **A** (a^* green-to-red opponent axis from CIE L*a*b*) channel, which captures the pink-to-green colour variation characteristic of monocyte cytoplasm — or a hierarchical approach that first distinguishes mononuclear from multi-lobed cells before refining within each group.

- **Generalisation across datasets.** All images were processed under similar staining and imaging conditions. Evaluation on external datasets such as LISC or BCCD would test whether the **H** (hue), **Cr** (red-difference chroma), and **Cb** (blue-difference chroma) ratio features generalise across microscopes, cameras, and staining protocols.

- **Platelet segmentation.** The current pipeline applies nucleus segmentation to platelet images, which do not contain nuclei. A pre-screening step that detects platelets before segmentation — for example, using the overall image size or the absence of a high-**S** (saturation — colour purity) large region — would be more principled and could improve robustness for edge cases.

- **Deep learning comparison.** A systematic comparison with fine-tuned CNN models on the same train/test split would contextualise these results within the broader state of the art.

- **Hyperparameter tuning for RF and KNN.** A thorough search comparable to that performed for the SVM might close the performance gap between models.

9 Conclusion

This project successfully implemented a complete, reproducible pipeline for the classification of blood cells into eight categories from peripheral blood smear images. The key contributions were:

1. A rotation-based augmentation strategy ($\pm 60^\circ$) that balanced the dataset from 17,092 imbalanced images to 28,000 balanced images (3,500 per class) while tracking provenance to prevent data leakage.

2. A mathematically grounded 51-feature extraction pipeline based on nucleus segmentation using the CMYK/HLS colour space combination, computing shape descriptors (Circularity, Convexity, Solidity) and ratio-normalised colour statistics across the NCL (nucleus), ROC (cytoplasm ring), and CVX (convex hull) regions in RGB, HSV, CIE L*a*b*, and YCrCb colour spaces.

3. A four-stage validation framework confirming mathematical validity, biological plausibility, inter-class separability (all 51 features with F-score ≥ 500), and benchmark accuracy before any model training.

4. A correlation and F-score based feature reduction compressing the feature space from 51 to 19 features with only 1.2 percentage points accuracy loss, retaining the most discriminative colour ratio features — dominated by **Cb** (blue-difference chroma), **Cr** (red-difference chroma), **R** (red channel), and **B** (blue channel) in the ROC cytoplasm region.

5. A two-stage SVM hyperparameter search (coarse random + fine grid) identifying optimal parameters ($C = 51{,}526.95$, $\gamma = 0.00101$, RBF kernel).

6. A systematic comparison of four classifiers under identical leakage-safe conditions, with the tuned SVM achieving the best performance: **CV accuracy 93.53% ± 0.58%**, **test accuracy 92.26%**, **macro F1 89.86%**.

The results demonstrate that carefully engineered classical features, when combined with a properly tuned SVM and a rigorous experimental protocol, can achieve strong classification performance on an eight-class blood cells problem without requiring GPU hardware or deep learning infrastructure.

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